

## Project Scope

| Project | Project Manager  | Date      |
|---------|------------------|-----------|
| Wadeeah | Munirah Aljrayan | 6/28/2026 |

| Justification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| Traditional clothing donation boxes are very basic and offer no protection against misuse. Unauthorized individuals can open them and take the donated clothes to resell for personal profit, which means the donations never reach the people they were intended for. This is a real problem that needs solving. This project addresses the problem by turning each donation box into a monitored unit that detects and reports any unauthorized opening or tampering in real time, ensuring donations are protected and reach those who genuinely need them. |

| Scope Description:<br>In Scope<br>Must (M), Should (S), Could (C), Won't (W)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| <ul style="list-style-type: none"><li>✓ Receive and record donations from boxes. Must (M).</li><li>✓ Generate an alert when an unauthorized attempt to open the box is detected. Must (M).</li><li>✓ Restrict box access to authorized users only, using RBAC for role-based access control. Must (M).</li><li>✓ Use JWT for authentication and session management. Must (M).</li><li>✓ Authenticate device messages using HMAC SHA-256. Must (M).</li><li>✓ Apply replay-prevention mechanisms (nonce / timestamp validation). Must (M).</li><li>✓ Audit every sensitive operation in a secure log. Must (M).</li><li>✓ Store user passwords securely using strong password hashing (BCrypt). Must (M).</li><li>✓ Provide a web dashboard that lists boxes, their status, and alerts. Must (M).</li><li>✓ Apply security monitoring and detection rules over the system logs. Must (M).</li><li>✓ Detect and notify when a box is full. Should (S).</li><li>✓ Provide incident-response playbooks for tamper, replay, and brute-force events. Should (S).</li><li>✓ Display a map view of box locations. Could (C).</li><li>✓ Provide basic reporting on box events over time. Could (C).</li></ul> |
| Out of Scope                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <ul style="list-style-type: none"><li>☒ Cash or payment handling (donations are clothes, not money). Won't (W).</li><li>☒ Physical IoT hardware (project uses a software simulator instead of a real device). Won't (W).</li><li>☒ Mobile application (web only). Won't (W).</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

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### Business Objectives

- Reduce unauthorized access to donation boxes by detecting and alerting on tampering attempts in real time, aiming to flag tamper events within seconds of detection.
- Increase trust for the charitable organization and its donors by ensuring donations are protected from theft and diversion.
- Enable timely collection by notifying staff when a box is full or tampered with, reducing the window where overflowing boxes are vulnerable.
- Deliver a secure, deployable prototype within the 50-day project timeline that demonstrates the full monitoring and security workflow.

### Project Deliverables

- Project objectives and scope
- Threat model
- Asset matrix and attack surface
- RBAC matrix
- Incident response playbook
- Donation box simulator
- Backend system (Spring Boot REST API)
- Web dashboard (Angular)
- Database
- Security design specifications (HMAC, JWT, replay prevention)
- Detection rules and monitoring setup
- Security testing and validation report
- Final security report
- Technical README and architecture diagram
- Source code repository (GitHub)
- Demo video

### Constraints

- The project must be completed within 50 days
- Work is part-time (3–4 hours/day)
- The team consists of only two students, each owning a separate track (development and security).
- Limited access to specialized security tools.
- No physical IoT hardware is available, so the donation box is represented by a **software simulator** rather than a real device.
- Development and testing run on local machines, not production-grade servers.
- Limited to the agreed feature set (the In-Scope list); anything marked "Won't" is excluded.
- No budget for paid services, licenses, cloud hosting, or hardware, limited to free and open-source tools.

## Project Scope

| Assumptions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| <p><b>Team Assumptions:</b></p> <ul style="list-style-type: none"><li>❖ Both team members will remain available and committed for the full 50 days.</li><li>❖ Both tracks (development and security) will progress in sync, so security testing can begin once each feature is built.</li></ul> <p><b>Technical Assumptions:</b></p> <ul style="list-style-type: none"><li>❖ The software simulator accurately represents how a real IoT donation box would behave, so results are realistic.</li><li>❖ The free and open-source tools chosen will be sufficient for the project's needs.</li></ul> <p><b>Scope Assumptions:</b></p> <ul style="list-style-type: none"><li>❖ The agreed requirements and scope will remain stable throughout the 50 days.</li></ul> |